UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER SECURITY
COURSE CODE	:	CSC662
EXAMINATION	:	OCTOBER 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC662 mock paper (October 2026). Computed answers show working; theory answers are supported by lecture slides (not lecturer-marked) with a verbatim quote and slide citation.

2. Supported theory marking points quote the lecture decks; any unresolved slot would remain a visible placeholder.

3. Part A = 10 x (2 + 6); Part B = 2 x (5 + 5); total = 100 marks.

4. Computed: 1 slot(s). Supported theory: 23 slots. Placeholders: 0.
Questions and answers are for local study use only.

5. Near-copy rejection used all 35 past papers; style widening is recorded per slot.

FOR MARKERS' USE ONLY

This marking scheme consists of 15 printed pages

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CSC662 - MARKING SCHEME (MOCK OCTOBER 2026)

Labels: computed / supported by lecture slides (not lecturer-marked) / placeholder. No official marking scheme exists for the real CSC662 papers; this is generated for the mock.

Entailment labels: supported = the quote directly supports the marking point (or the point passed the stem-overlap gate without an LLM check); partly supported = the quote supports part of the point or is ambiguous; not supported = the quote does not support the point. Only 'not supported' points triggered regeneration.

Part A: 10 x (2 + 6) = 80 marks. Part B: 2 x (5 + 5) = 20 marks. Total: 100 marks. Slots: 1 computed, 23 supported by lecture slides, 0 placeholder.

Methods: computed answers use verified solvers. Theory answers are supported by lecture slides (not lecturer-marked). Each marking point carries a verbatim quote, a deck/slide/chunk citation and a term-overlap score; low overlap is flagged for human review. Sources are restricted to the slot topic's allowed decks, and a slide is not reused across questions where an alternative exists.

PART A - QUESTION 1

A1a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

List TWO (2) differences between morals and laws as described in the lecture on ethics and computer security.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Morals are described as traditions of belief concerning right and wrong conduct.	"Morals are traditions of belief about right and wrong conduct"	071 - Ethics and Computer Security, slide 5	071 - Ethics and Computer Security::s05::1	0.86	supported (ste gate)
2	Laws are described as formal rules of conduct imposed by a sovereign authority such as a government.	"Laws are formal rules of conduct that a sovereign authority, such as a government, imposes on its subjects or citizens"	071 - Ethics and Computer Security, slide 5	071 - Ethics and Computer Security::s05::1	1.00	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 071 - Ethics and Computer Security, slide 5

A1b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Discuss how a professional code of ethics and conduct guides a computing professional in carrying out security-related work. Support your discussion with six (6) distinct obligations drawn from the ACM code.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The professional must aim for high quality in the processes and products of their professional work.	"Strive to achieve high quality in both the processes and products of professional work."	071 - Ethics and Computer Security, slide 12	071 - Ethics and Computer Security::s12::1	0.75	supported (ste gate)
2	The professional must uphold strong levels of competence, conduct and ethical practice.	"Maintain high standards of professional competence, conduct, and ethical practice."	071 - Ethics and Computer Security, slide 12	071 - Ethics and Computer Security::s12::1	0.56	supported (ste gate)
3	The professional must give thorough evaluations of computer systems and their impacts, including possible risks.	"Give comprehensive and thorough evaluations of computer systems and their impacts, including analysis of possible risks."	071 - Ethics and Computer Security, slide 12	071 - Ethics and Computer Security::s12::1	0.80	supported (ste gate)
4	The professional must only access computing	"Access computing and communication resources only	071 - Ethics and Computer	071 - Ethics and Computer	0.75	supported (ste gate)

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	and communication resources when authorised or compelled by the public good.	when authorized or when compelled by the public good."	Security, slide 12	Security::s12::1		
5	The professional must respect privacy as part of the code of ethics.	"Respect privacy."	071 - Ethics and Computer Security, slide 11	071 - Ethics and Computer Security::s11::1	0.29 LOW	supported
6	The professional must honour confidentiality as part of the code of ethics.	"Honor confidentiality."	071 - Ethics and Computer Security, slide 11	071 - Ethics and Computer Security::s11::1	0.14 LOW	supported

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 071 - Ethics and Computer Security, slide 11; 071 - Ethics and Computer Security, slide 12

PART A - QUESTION 2

A2a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain how a substitution cipher works, stating what serves as the key and which of Shannon's principles it corresponds to.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The cipher works by replacing each plaintext letter with a different letter, producing the ciphertext.	"In substitution ciphers, we replace the plain-text letters with other letters"	031 - Classical Cryptography, slide 4	031 - Classical Cryptography::s04::1	0.43 LOW	supported
2	The substitution rule itself acts as the key, and this corresponds to Shannon's principle of confusion.	"The substitution rule is the key"	031 - Classical Cryptography, slide 4	031 - Classical Cryptography::s04::1	0.25 LOW	partly supporte

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 031 - Classical Cryptography, slide 4

A2b - 6 marks - computed - matches lecture slide 15 (standard tableau)

Question:

Decrypt the following ciphertext using the Vigenere cipher with the key CIPHER. Use the standard 26 x 26 Vigenere tableau, writing the key over the message and repeating it as necessary.
UMRBVVNWVPREQE

Answer: SECURELOGINNOW

The 26 x 26 Vigenere tableau is printed in the question paper.

Working:

```
Standard 26 x 26 tableau, A=0; key repeated over the ciphertext.
Ciphertext: UMRBVVNWVPREQE
Key:        CIPHERCIPHERCI
Letter-by-letter: C - K mod 26 = P
  U (20) - C ( 2) = 18 -> S
  M (12) - I ( 8) =  4 -> E
  R (17) - P (15) =  2 -> C
  B ( 1) - H ( 7) = 20 -> U
  V (21) - E ( 4) = 17 -> R
  V (21) - R (17) =  4 -> E
  N (13) - C ( 2) = 11 -> L
  W (22) - I ( 8) = 14 -> O
  V (21) - P (15) =  6 -> G
  P (15) - H ( 7) =  8 -> I
  R (17) - E ( 4) = 13 -> N
  E ( 4) - R (17) = 13 -> N
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  Q (16) - C ( 2) = 14 -> O
  E ( 4) - I ( 8) = 22 -> W
Plaintext:  SECURELOGINNOW
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Suggested marks: 1 mark for the key/table setup; 4 marks for the letter-by-letter working; 1 mark for the complete answer.

Confidence: computed - matches lecture slide 15 (standard tableau).

PART A - QUESTION 3

A3a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain the Chinese Wall model as a formal security model, stating the concern it addresses and how its access decisions differ from static policies.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The Chinese Wall policy was proposed by Brewer and Nash to address conflicts of interest.	"Brewer and Nash (1989) proposed a policy called the Chinese Wall Policy that addresses such conflicts of interest."	024 - Formal Security Model, slide 21	024 - Formal Security Model::s21::1	1.00	supported (ste gate)
2	Access decisions depend on the subject's history of past accesses, so permissions change dynamically.	"The Chinese Wall is an access control policy in which accesses are sensitive to the history of past accesses."	024 - Formal Security Model, slide 27	024 - Formal Security Model::s27::1	0.33 LOW	supported

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.

Citations: 024 - Formal Security Model, slide 21; 024 - Formal Security Model, slide 27

A3b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Describe the access control matrix, the access control list and the capability list as access control structures, giving for each structure one advantage and one disadvantage, and state two requirements that any access control structure must satisfy.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The access control matrix is identified as a basic control structure whose advantages are clarity of definition and ease of verification.	"ACM is a basic control structure."	023 - Access Control and Authorization, slide 8	023 - Access Control and Authorization::s08::1	0.27 LOW	partly supporte
2	The access control matrix is noted to scale poorly and to handle changes poorly.	"poor scalability poor handling of changes"	023 - Access Control and Authorization, slide 8	023 - Access Control and Authorization::s08::1	0.38 LOW	supported
3	An ACL is described as an object-centred description of access rights, which gives easy access to the access rights of an object.	"ACL is an object-centered description of access rights:"	023 - Access Control and Authorization, slide 10	023 - Access Control and Authorization::s10::1	0.71	supported (ste gate)
4	The ACL approach is criticised for giving a poor overview of access rights per subject and for making revocation difficult.	"poor overview of access rights per subject difficulty of revocation"	023 - Access Control and Authorization, slide 10	023 - Access Control and Authorization::s10::1	0.58	supported (ste gate)
5	Capability lists are named as one of the mechanisms for implementing access policies, alongside the	"access control matrix (ACM) Capabilities (C-List) access control lists (ACL)"	023 - Access Control and Authorization, slide 7	023 - Access Control and Authorization::s07::1	0.56	supported (ste gate)

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	access control matrix and ACLs.					
6	A required property of access control structures is the ability to express control policies together with verifiability of correctness.	"an ability to express control policies verifiability of correctness."	023 - Access Control and Authorization, slide 7	023 - Access Control and Authorization::s07::1	0.46 LOW	supported

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 023 - Access Control and Authorization, slide 10; 023 - Access Control and Authorization, slide 7; 023 - Access Control and Authorization, slide 8

PART A - QUESTION 4

A4a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain two characteristics of Trojan horse malware, including how it disguises itself and what harm it can cause once active.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Trojans disguise themselves by appearing to be useful or benign software.	"They usually masquerade as helpful or harmless programs"	051 - Software Security, slide 12	051 - Software Security::s12::1	0.00 LOW	supported
2	Once running, a Trojan can steal private data, delete files, log keystrokes, or open a way for further malware.	"begin to steal private data, delete files, log keystrokes or provide an opening for further malware to get installed"	051 - Software Security, slide 12	051 - Software Security::s12::1	0.67	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 051 - Software Security, slide 12

A4b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

State six techniques or behaviours by which malware propagates itself or conceals its activity on an infected system, drawing on the historical and technical examples covered in the course.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	A virus can spread by infecting the boot sector of a disk, as with Elk Cloner which spread via boot sector infection.	"A program spread via boot sector infection"	051 - Software Security, slide 7	051 - Software Security::s07::1	0.50	supported (ste gate)
2	A worm can propagate across a network by exploiting a remote execution feature of a service, as the Morris worm did with sendmail.	"Used a debugging feature of sendmail (remote execution)"	051 - Software Security, slide 7	051 - Software Security::s07::1	0.25 LOW	supported
3	A virus can append its body to a program and redirect the program entry point to the virus code, restoring the original entry point afterwards.	"Replace a program entry point with a jump to a virus body"	051 - Software Security, slide 9	051 - Software Security::s09::1	0.42 LOW	partly supporte
4	A virus can overwrite a program header so that	"Over write a program header to switch the entry point to a virus"	051 - Software Security, slide	051 - Software Security::s10::1	0.46 LOW	partly supporte

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	the entry point switches to the virus, then jump to the original entry point during execution.		10			
5	A rootkit hides malicious processes and software so that the infected machine stays under the author's remote control, and it is hard to detect because it hides from detection software.	"Rootkits are a type of malware that mask malicious processes and software in the infected systems to ensure privileged remote access."	051 - Software Security, slide 13	051 - Software Security::s13::1	0.40 LOW	partly supporte
6	Spyware secretly collects user information such as logins, passwords, visited sites and keystrokes, and can redirect the browser to unwanted sites.	"They steal user logins and passwords, document sites visited, log keystrokes and even redirect browsers to unwanted sites."	051 - Software Security, slide 13	051 - Software Security::s13::1	0.60	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 051 - Software Security, slide 10; 051 - Software Security, slide 13; 051 - Software Security, slide 7; 051 - Software Security, slide 9

PART A - QUESTION 5

A5a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

State TWO (2) typical locations where network-based IDS sensors should be placed to protect an organization's most valuable assets.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	A sensor can be placed just inside the firewall.	"Just inside the firewall"	062 - Intrusion Detection System, slide 7	062 - Intrusion Detection System::s07::1	0.50	supported (ste gate)
2	A sensor can be placed on the DMZ.	"On the DMZ"	062 - Intrusion Detection System, slide 7	062 - Intrusion Detection System::s07::1	0.25 LOW	supported

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 062 - Intrusion Detection System, slide 7

A5b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Refer to your answer in (a). Based on the scenario in (a), explain how the organization can strengthen its preventive network security mechanisms by describing the role of a firewall, a proxy server, and network segmentation in controlling and protecting network traffic.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	A firewall controls incoming and outgoing network traffic according to predetermined security policies.	"Control incoming and outgoing traffic on network with predetermined security policies."	061 - Network Security, slide 3	061 - Network Security::s03::1	0.80	supported (ste gate)
2	A proxy server improves data security and network performance by acting as a firewall and web	"Proxy servers act as a firewall and web filter, provide shared network connections, and cache data to speed up common requests."	061 - Network Security, slide 3	061 - Network Security::s03::1	0.73	supported (ste gate)

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	filter.					
3	Network segmentation creates boundaries between segments whose computing assets share a common function, risk, or role.	"Boundaries between network segments where computing assets within the segment have a common function, risk or role within an organization."	061 - Network Security, slide 3	061 - Network Security::s03::1	0.77	supported (ste gate)
4	A remote access VPN gives individual hosts or clients secure remote access to the organization's network.	"Provides remote and secure access to an organization network to individual hosts or clients"	061 - Network Security, slide 4	061 - Network Security::s04::1	0.89	supported (ste gate)
5	Zero Trust Network Access grants a user only the access and permissions needed to fulfill the role.	"A user should only has access and permission that required to fulfill the role"	061 - Network Security, slide 4	061 - Network Security::s04::1	0.55	supported (ste gate)
6	IDS/IPS complement these preventive mechanisms by detecting or preventing attacks such as brute force, Denial-of-Service, and exploits of network vulnerabilities.	"To detect or prevent the network security attacks such as brute force attacks, Denial-of-Service attacks and exploits of network vulnerabilities."	061 - Network Security, slide 4	061 - Network Security::s04::1	0.73	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 061 - Network Security, slide 3; 061 - Network Security, slide 4

PART A - QUESTION 6

A6a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain two ways in which a documented code of ethics shapes the ethical conduct of computing professionals in security-related work.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	It sets out a written record of which behaviours are acceptable for the professional or social group.	"Documented set of acceptable behaviors for professional or social groups."	071 - Ethics and Computer Security, slide 6	071 - Ethics and Computer Security::s06::1	0.56	supported (ste gate)
2	It makes clearer what the accepted standards of behaviour are.	"Adds clarity to the understanding of acceptable standards of behavior."	071 - Ethics and Computer Security, slide 8	071 - Ethics and Computer Security::s08::1	0.40 LOW	supported

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 071 - Ethics and Computer Security, slide 6; 071 - Ethics and Computer Security, slide 8

A6b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain six responsibilities that a computing professional owes to users and the organisation when handling private data in security-related work, drawing on the lecture's discussion of ethics and professional conduct.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Recognise that users and information specialists carry definite duties to act within ethical, moral and legal limits when handling data.	"Users in general, and information specialists in particular, have definite responsibilities in terms of performing within ethical, moral, and legal constraints"	071 - Ethics and Computer Security, slide 3	071 - Ethics and Computer Security::s03::1	0.50	supported (ste gate)

2	Accept that merely describing the moral and ethical dimensions of ICT is insufficient for responsible practice.	"Simply describing may not be enough when discussing the moral and ethical dimensions of ICT"	071 - Ethics and Computer Security, slide 3	071 - Ethics and Computer Security::s03::1	0.44 LOW	supported
3	Follow the code of ethics and professional conduct published by the Association of Computing Machinery.	"Association of Computing Machinery ACM (founded 1947) has a code of ethics and professional conduct."	071 - Ethics and Computer Security, slide 7	071 - Ethics and Computer Security::s07::1	0.78	supported (ste gate)
4	Recognise that professional bodies such as the ACM publish a code of ethics and professional conduct that practitioners can follow.	"Association of Computing Machinery ACM (founded 1947) has a code of ethics and professional conduct."	071 - Ethics and Computer Security, slide 7	071 - Ethics and Computer Security::s07::1	0.42 LOW	supported
5	Consult the IEEE Computer Society code as a further professional standard governing conduct.	"Computer Society of the Institute of Electrical and Electronics Engineers (IEEE - CS) (founded in 1946)."	071 - Ethics and Computer Security, slide 7	071 - Ethics and Computer Security::s07::1	0.30 LOW	partly supporte
6	Recognise that professional bodies such as the ACM, AITP, IEEE-CS and PMI each publish codes of ethics and professional conduct that practitioners can follow.	"Association of Computing Machinery ACM (founded 1947) has a code of ethics and professional conduct."	071 - Ethics and Computer Security, slide 7	071 - Ethics and Computer Security::s07::1	0.33 LOW	partly supporte

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 071 - Ethics and Computer Security, slide 3; 071 - Ethics and Computer Security, slide 7

PART A - QUESTION 7

A7a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

List two classes of threats described in the course and, for each, state the security objective that is affected and the type of harm it represents.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Disclosure is a threat class involving unauthorized access to information, which harms confidentiality.	"Disclosure - unauthorized access to information"	011 - Introduction to Computer Security, slide 10	011 - Introduction to Computer Security::s10::1	0.44 LOW	partly supporte
2	Deception is a threat class involving acceptance of false data, which harms integrity.	"Deception - acceptance of false data"	011 - Introduction to Computer Security, slide 10	011 - Introduction to Computer Security::s10::1	0.44 LOW	partly supporte

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 011 - Introduction to Computer Security, slide 10

A7b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Describe the Samy worm as a case study of a cross-site scripting attack. In your answer, explain how the malicious script was injected, how it spread between users, how the victim's browser was made to act, and the scale of the resulting infection.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The attacker injected JavaScript into a user profile by writing to document.body.innerHTML.	"Inject Javascript into a user profile using document.body.innerHTML"	052 - Web Security, slide 15	052 - Web Security::s15::1	0.70	supported (ste gate)
2	The script obtained the list of the user's heroes by issuing a GET request on a user profile.	"Get the list of user's heros by using a GET on a user profile"	052 - Web Security, slide 15	052 - Web Security::s15::1	0.38 LOW	supported
3	The script added Samy as a friend by performing an XML-HTTP request on the addFriends page.	"Add Samy as a field by performing XML-HTTP on the addFriends page"	052 - Web Security, slide 15	052 - Web Security::s15::1	0.70	supported (ste gate)
4	Any user visiting an infected profile had their own profile and friends list infected, so the worm spread from visitor to visitor.	"Any user who visits an infected user's page will gets profile infected with Javascript and his friends list infected with Samy"	052 - Web Security, slide 15	052 - Web Security::s15::1	0.46 LOW	supported
5	The victim's browser executed the malicious script contained in the response, sending the victim's cookies to the attacker's server.	"The victim's browser executes the malicious script inside the response, sending the victim's cookies to the attacker's server."	052 - Web Security, slide 16	052 - Web Security::s16::1	0.91	supported (ste gate)
6	The infection eventually reached about 1,000,000 MySpace profiles.	"Eventually about 1,000,000 MySpace profiles were infected."	052 - Web Security, slide 15	052 - Web Security::s15::1	0.71	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 052 - Web Security, slide 15; 052 - Web Security, slide 16

PART A - QUESTION 8

A8a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

State the two stages of the biometric authentication process and, for each stage, name the activities it involves.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Identification stage: acquisition, creation of master characteristics, and storage of master characteristics.	"Identification Stage Acquisition Creation of Master characteristics Storage of Master characteristics"	072 - Biometrics and Authentication, slide 5	072 - Biometrics and Authentication::s05::1	1.00	supported (ste gate)
2	Authentication stage: acquisition(s), comparison, and decision.	"Authentication Stage Acquisition(s) Comparison Decision"	072 - Biometrics and Authentication, slide 5	072 - Biometrics and Authentication::s05::1	1.00	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 072 - Biometrics and Authentication, slide 5

A8b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain the categories of biometric applications and the characteristics of the biometric technologies used for authentication, giving examples of each category and the practical issues associated with these technologies.

Marking points:

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No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Biometric applications are divided into two sectors, namely physiological and behavioural.	"Biometric applications available today are categorized into 2 sectors"	072 - Biometrics and Authentication, slide 4	072 - Biometrics and Authentication::s04::1	0.57	supported (ste gate)
2	The physiological category includes iris, fingerprints, hand, retinal and face recognition.	"Psychological: Iris, Fingerprints, Hand, Retinal and Face recognition"	072 - Biometrics and Authentication, slide 4	072 - Biometrics and Authentication::s04::1	0.67	supported (ste gate)
3	The behavioural category includes voice, typing pattern and signature.	"Behavioural: Voice, Typing pattern, Signature"	072 - Biometrics and Authentication, slide 4	072 - Biometrics and Authentication::s04::1	0.67	supported (ste gate)
4	A biometric is defined as a physiological or behavioural characteristic that can distinguish one person from another.	"A biometric is a physiological or behavioural characteristic of a human being that can distinguish one person from another"	072 - Biometrics and Authentication, slide 2	072 - Biometrics and Authentication::s02::1	0.88	supported (ste gate)
5	Iris scanning requires no touch and overcomes the issues of retinal scanners.	"No touch required"	072 - Biometrics and Authentication, slide 10	072 - Biometrics and Authentication::s10::1	0.25 LOW	partly supporte
6	Fingerprint biometrics suffer from disadvantages such as racial issues, dirt, grime and wounds, and can be spoofed so liveness is important.	"Can be spoofed - liveness important!"	072 - Biometrics and Authentication, slide 7	072 - Biometrics and Authentication::s07::1	0.29 LOW	partly supporte

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 072 - Biometrics and Authentication, slide 10; 072 - Biometrics and Authentication, slide 2; 072 - Biometrics and Authentication, slide 4; 072 - Biometrics and Authentication, slide 7

PART A - QUESTION 9

A9a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

State two ways in which an intrusion detection system contributes to detecting and defending against attacks on a system.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	An IDS can confirm whether an attack succeeded or failed by examining the log entries recorded by a host-based IDS.	"Verifies success or failure of attack by reviewing HIDS log entries"	062 - Intrusion Detection System, slide 14	062 - Intrusion Detection System::s14::1	0.23 LOW	supported
2	An IDS can provide detection and response to an attack in near real time.	"Near real-time detection and response"	062 - Intrusion Detection System, slide 14	062 - Intrusion Detection System::s14::1	0.56	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 062 - Intrusion Detection System, slide 14

A9b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

State six points that distinguish the three security concepts of vulnerability, threat and attack, and give one countermeasure example for a threat class, drawing on the course material.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	A vulnerability is defined as a weakness of a system that can be exploited to damage assets.	"Vulnerabilities: Weakness of a system that can be exploited to damage assets"	011 - Introduction to Computer Security, slide 9	011 - Introduction to Computer Security::s09::1	0.88	supported (ste gate)
2	A threat is described as a potential violation of security rather than an actual one.	"Threats: A threat is a "potential" violation of security"	011 - Introduction to Computer Security, slide 9	011 - Introduction to Computer Security::s09::1	0.57	supported (ste gate)
3	An attack is the actual violation of security, distinguishing it from a mere threat.	"Attacks: The actual violation of security"	011 - Introduction to Computer Security, slide 9	011 - Introduction to Computer Security::s09::1	0.57	supported (ste gate)
4	A threat is an object, person or other entity that represents a constant danger to an asset.	"Threat: object, person, or other entity representing a constant danger to an asset"	011 - Introduction to Computer Security, slide 28	011 - Introduction to Computer Security::s28::1	1.00	supported (ste gate)
5	An attack is a deliberate act that exploits a vulnerability.	"Attack: a deliberate act that exploits vulnerability"	011 - Introduction to Computer Security, slide 28	011 - Introduction to Computer Security::s28::1	1.00	supported (ste gate)
6	Sandboxing is a countermeasure providing a safe, protected and isolated environment to run or open suspected insecure programs or files.	"Sandboxing A safe, protected and isolated environment in computer system to run or open suspected insecure programs or files."	061 - Network Security, slide 26	061 - Network Security::s26::1	0.85	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 011 - Introduction to Computer Security, slide 28; 011 - Introduction to Computer Security, slide 9; 061 - Network Security, slide 26

PART A - QUESTION 10

A10a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain how a buffer overflow arises in a program such as the one using strcpy() into a fixed-size local buffer, and state what an attacker can achieve by exploiting it.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The overflow occurs because strcpy() copies the input string without checking its length against the size of the destination buffer.	"strcpy() do not check the string length"	051 - Software Security, slide 23	051 - Software Security::s23::1	0.29 LOW	supported
2	By controlling the overflowing buffer content, the attacker can overwrite the saved return address and thereby hijack the program's execution flow.	"By controlling buffer content, an attacker can overwrite the return address with an arbitrary value and hijack the execution flow."	051 - Software Security, slide 21	051 - Software Security::s21::1	0.67	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 051 - Software Security, slide 21; 051 - Software Security, slide 23

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A10b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Windows is often described as offering rich but complicated protection facilities. Explain six aspects of Windows security mechanisms, covering how subjects and their credentials are handled, how resources and policies are organised, and why the complexity of these mechanisms can itself become a security problem.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Identify the active entities in Windows as processes and threads.	"Windows subjects are processes and threads."	041 - Windows Security, slide 11	041 - Windows Security::s11::1	0.60	supported (ste gate)
2	State where the security credentials of a subject are kept.	"Security credentials for a subject are stored in a token."	041 - Windows Security, slide 11	041 - Windows Security::s11::1	0.60	supported (ste gate)
3	Describe the directory service introduced in Windows 2000 that organises users, groups, credentials, resources and policies.	"Active directory introduced in Windows 2000 is an LDAP-like directory service for organization of system resources:"	041 - Windows Security, slide 6	041 - Windows Security::s06::1	0.55	supported (ste gate)
4	Name the hierarchical database holding critical system information such as the SAM database and security logs.	"A hierarchical database containing critical system information"	041 - Windows Security, slide 4	041 - Windows Security::s04::1	0.50	supported (ste gate)
5	Note that discretionary access control allows fine-grained control over many different kinds of entities, both in Windows and in applications.	"DACS enable fine-grained access control of heterogeneous entities in Windows itself as well as applications."	041 - Windows Security, slide 21	041 - Windows Security::s21::1	0.43 LOW	supported
6	Explain that the very complexity of Windows security can backfire, since bad configuration or coding bugs can cause serious security incidents.	"The complexity of Windows security mechanisms is also its enemy: misconfiguration as well as implementation bugs may lead to severe security incidents."	041 - Windows Security, slide 21	041 - Windows Security::s21::1	0.36 LOW	supported

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 041 - Windows Security, slide 11; 041 - Windows Security, slide 21; 041 - Windows Security, slide 4; 041 - Windows Security, slide 6

PART B - QUESTION 1

B1a - 5 marks - supported by lecture slides (not lecturer-marked)

Scenario:

A small company runs a shared Linux server where staff log in with user names and passwords. After several accounts were guessed by an attacker, the administrator is asked to strengthen how passwords are protected and managed on the system.

Question:

Explain five password security controls that the administrator can apply to protect the accounts on this server, and state how passwords should be stored.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Apply access control to the password files so that they cannot be read freely.	"Access control applied to password files"	022 - Identification and Authentication, slide 12	022 - Identification and Authentication::s12::1	0.44 LOW	supported
2	De-activate an account after multiple failed	"De-activation of accounts after multiple failed logins"	022 - Identification	022 - Identification and	0.86	supported (ste gate)

	login attempts.		and Authentication, slide 12	Authentication::s12::1		
3	Display the time of the last login so users can notice unexpected access.	"Displaying the time of a last login"	022 - Identification and Authentication, slide 12	022 - Identification and Authentication::s12::1	0.44 LOW	partly supporte
4	Enforce password policies, for example for the root password.	"Password policies, e.g. for root password"	022 - Identification and Authentication, slide 12	022 - Identification and Authentication::s12::1	0.60	supported (ste gate)
5	Store the password in encrypted form using a one-way hash function.	"Password is stored in an encrypted form using a one-way hash function:"	022 - Identification and Authentication, slide 8	022 - Identification and Authentication::s08::1	1.00	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 022 - Identification and Authentication, slide 12; 022 - Identification and Authentication, slide 8

B1b - 5 marks - supported by lecture slides (not lecturer-marked)

Question:

Discuss how single sign-on and Kerberos could reduce the password burden on the staff of the company in the scenario, and explain the design criteria that make Kerberos suitable for authenticating users across its distributed servers.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Single sign-on reduces the number of passwords users must remember.	"Reduce the need to memorize passwords"	022 - Identification and Authentication, slide 16	022 - Identification and Authentication::s16::1	0.25 LOW	supported
2	Single sign-on reduces the effort needed to maintain passwords.	"Reducing password maintenance effort"	022 - Identification and Authentication, slide 16	022 - Identification and Authentication::s16::1	0.43 LOW	supported
3	Kerberos is a client-server single sign-on architecture that automates authentication in large distributed environments.	"Single sign-on methods such as Kerberos provide means for automating authentication in large distributed environments."	022 - Identification and Authentication, slide 24	022 - Identification and Authentication::s24::1	0.73	supported (ste gate)
4	Kerberos is designed so that passwords are never sent over the network in clear text, protecting against eavesdropping.	"Security against eavesdropping: no password transmission in clear text."	022 - Identification and Authentication, slide 18	022 - Identification and Authentication::s18::1	0.42 LOW	supported
5	Kerberos is transparent, so after the initial login the user is unaware of further authentication.	"Transparency: user is not aware of any authentication beyond an initial login."	022 - Identification and Authentication, slide 18	022 - Identification and Authentication::s18::1	0.56	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 022 - Identification and Authentication, slide 16; 022 - Identification and Authentication, slide 18; 022 - Identification and Authentication, slide 24

PART B - QUESTION 2

B2a - 5 marks - supported by lecture slides (not lecturer-marked)

Scenario:

A hospital wants to prove to auditors that every server booted only approved software. The vendor proposes placing a tamper-resistant chip on each motherboard and building a chain of trust from that chip upward through the boot software.

Question:

Referring to the hospital's proposed tamper-resistant chip, explain how a chain of trust is established from the hardware up to the running platform, and state what the chip is responsible for.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The chain of trust begins at the hardware TPM and then proceeds through the software stages (RTM, TPM software stack, BIOS, kernel).	"The chain starts with the hardware TPM, then software:"	073 - Trusted Computing, slide 8	073 - Trusted Computing::s08::1	0.36 LOW	partly supporte
2	At each stage, the current stage measures and validates the next one.	"Each stage measures and validates the next one"	073 - Trusted Computing, slide 8	073 - Trusted Computing::s08::1	0.80	supported (ste gate)
3	The resulting measurements are recorded in Platform Configuration Registers (PCRs) on the TPM.	"Measurements go into Platform Configuration Registers (PCRs) on the TPM"	073 - Trusted Computing, slide 8	073 - Trusted Computing::s08::1	0.75	supported (ste gate)
4	The TPM is a tamper-resistant hardware module mounted in the platform.	"Trusted Platform Module (TPM): a tamper-resistant hardware module mounted in a platform."	073 - Trusted Computing, slide 6	073 - Trusted Computing::s06::1	1.00	supported (ste gate)
5	The TPM is responsible for measurement, protected storage, integrity reporting and policy enforcement.	"Responsible for: measurement, (protected) storage, (integrity) reporting and policy enforcement."	073 - Trusted Computing, slide 6	073 - Trusted Computing::s06::1	0.89	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 073 - Trusted Computing, slide 6; 073 - Trusted Computing, slide 8

B2b - 5 marks - supported by lecture slides (not lecturer-marked)

Question:

The vendor's tamper-resistant chip is said to support NGSCB. State five things about the Nexus and the Nexus Computing Agent that the hospital's security team should understand before relying on this design.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Nexus is a security kernel that is authenticated when the platform boots.	"Security kernel, authenticated on boot"	073 - Trusted Computing, slide 14	073 - Trusted Computing::s14::1	0.57	supported (ste gate)
2	Nexus provides the application interface to the TPM.	"Application interface to TPM"	073 - Trusted Computing, slide 14	073 - Trusted Computing::s14::1	0.60	supported (ste gate)
3	Nexus does not place trust in the operating system.	"Does not trust OS"	073 - Trusted Computing, slide 14	073 - Trusted Computing::s14::1	0.43 LOW	supported
4	The Nexus Computing Agent is a trusted software component that runs in trusted mode and communicates with Nexus.	"Runs in trusted mode that communicates with Nexus"	073 - Trusted Computing, slide 15	073 - Trusted Computing::s15::1	0.62	supported (ste gate)
5	Developers are able to create their own agents	"Developers can make their own agents to run on the trusted	073 - Trusted Computing,	073 - Trusted Computing::s15::	0.75	supported (ste gate)

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	to run on the trusted platform.	platform"	slide 15	1		

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 073 - Trusted Computing, slide 14; 073 - Trusted Computing, slide 15

End of marking scheme.

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